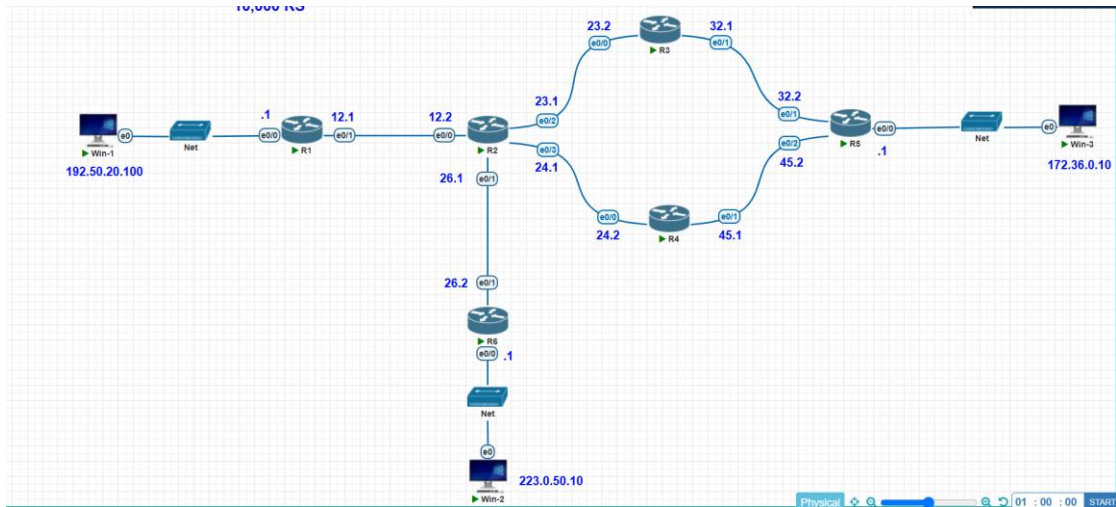


Multi-Router Network Lab

Network Topology



Win-1

1. Open ncpa.cpl.
2. Right-click Local Area Connection fi Properties.
3. Select Internet Protocol (TCP/IP) fi Properties.
4. Select Use the following IP address.
5. Enter IP 192.50.20.100, mask 255.255.255.0, gateway 192.50.20.1.
6. Click OK and close the dialogs.

Win-2

Repeat the same procedure and enter IP 223.0.50.10, mask 255.255.255.0, gateway 223.0.50.1.

Win-3

Repeat the same procedure and enter IP 172.36.0.10, mask 255.255.0.0, gateway 172.36.0.1.

Verify the Windows configuration

Open Command Prompt and run:

Ipconfig

Ping <default gateway-IP>

The gateway ping should return replies before testing remote networks.

Router Configuration

R1

```
interface e0/0
ip address 192.50.20.1 255.255.255.0 .0
no shutdown
interface e0/1
ip address 12.0.0.1 255.0.0.0
no shutdown
```

R2

```
interface e0/0
ip address 12.0.0.2 255.0.0.0
no shutdown
interface e0/2
ip address 23.0.0.1 255.0.0.0
no shutdown
interface e0/3
ip address 24.0.0.1 255.0.0.0
no shutdown
interface e0/1
ip address 26.0.0.1 255.0.0.0
no shutdown
```

R3

```
interface e0/0
ip address 23.0.0.2 255.0.0.0
no shutdown
interface e0/1
ip address 32.0.0.1 255.0.0.0
no shutdown
```

R4

```
interface e0/0
ip address 24.0.0.2 255.0.0.0
no shutdown
interface e0/1
ip address 45.0.0.1 255.0.0.0
no shutdown
```

R5

```
interface e0/1
ip address 32.0.0.2 255.0.0.0
no shutdown
interface e0/2
ip address 45.0.0.2 255.0.0.0
no shutdown
interface e0/0
ip address 172.36.0.1 255.255.0.0
no shutdown
```

R6

```
interface e0/1
ip address 26.0.0.2 255.255.255.0
no shutdown
interface e0/0
ip address 223.0.50.1 255.255.255.0
no shutdown
```

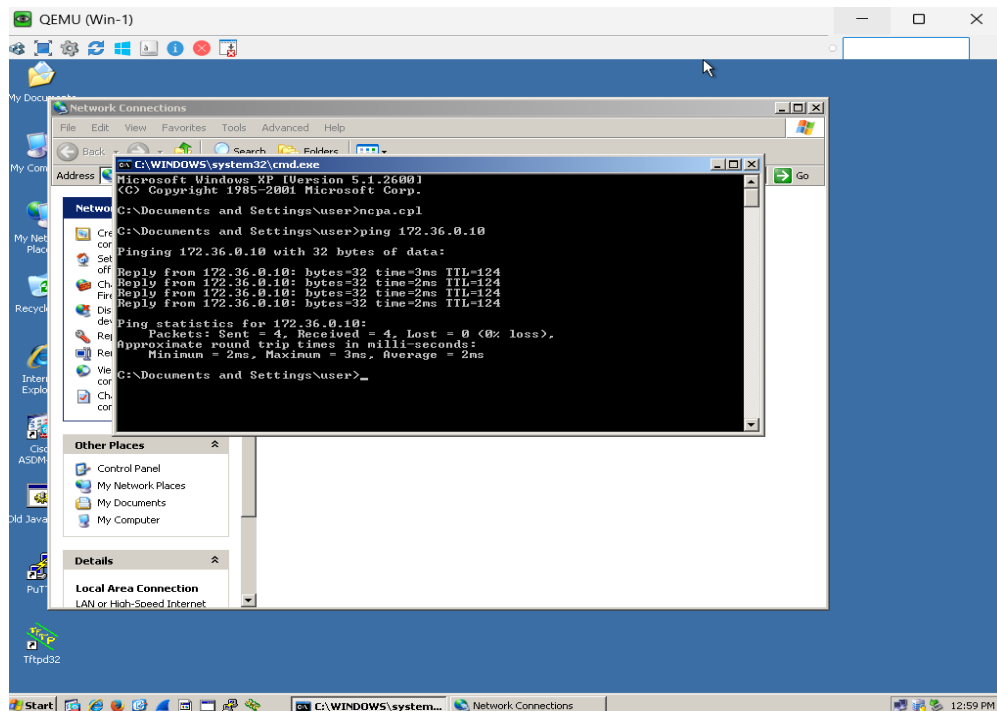
Verify Interfaces and Routing

Run the following commands on the routers:

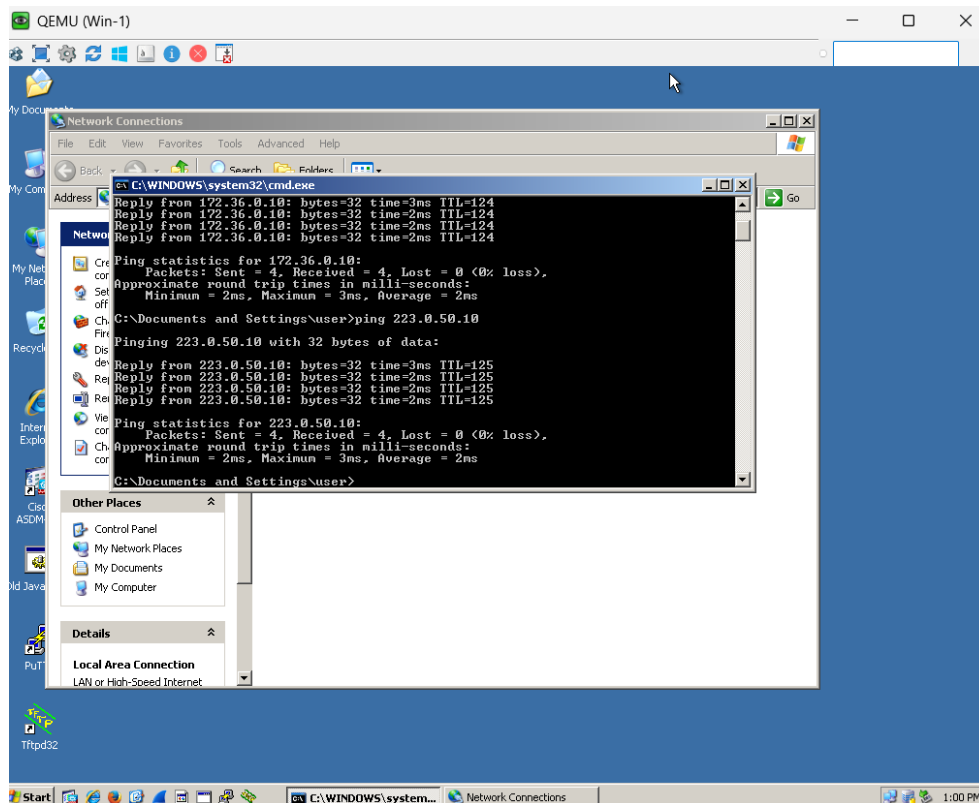
```
show ip interface brief
show ip route
ping <destination-IP>
```

Ping Verification Screenshots

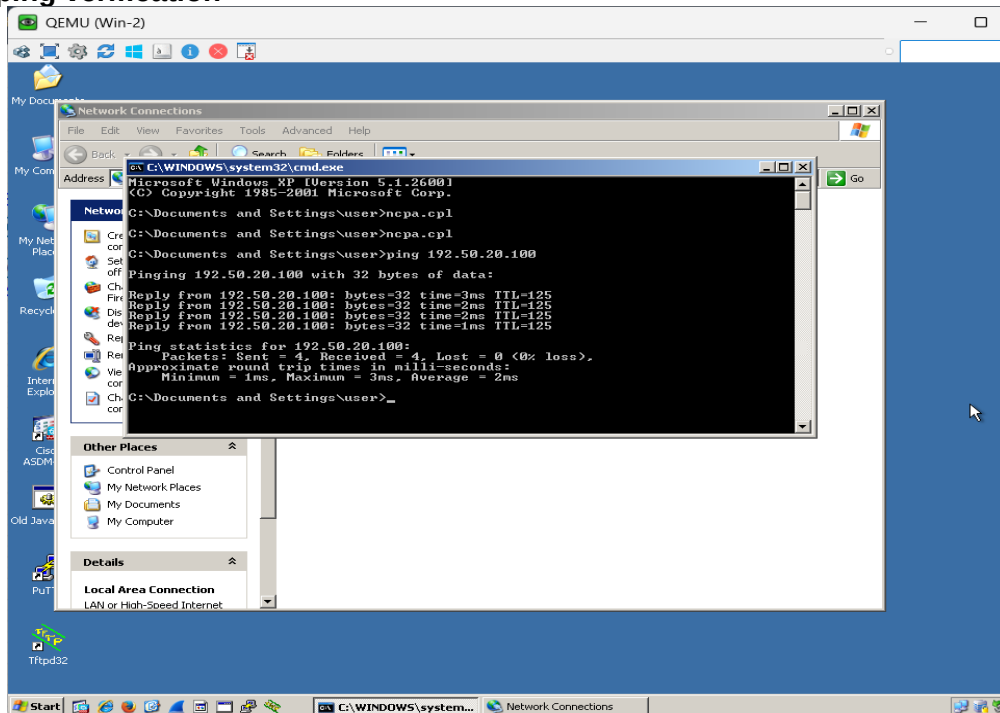
Win-1 ping verification



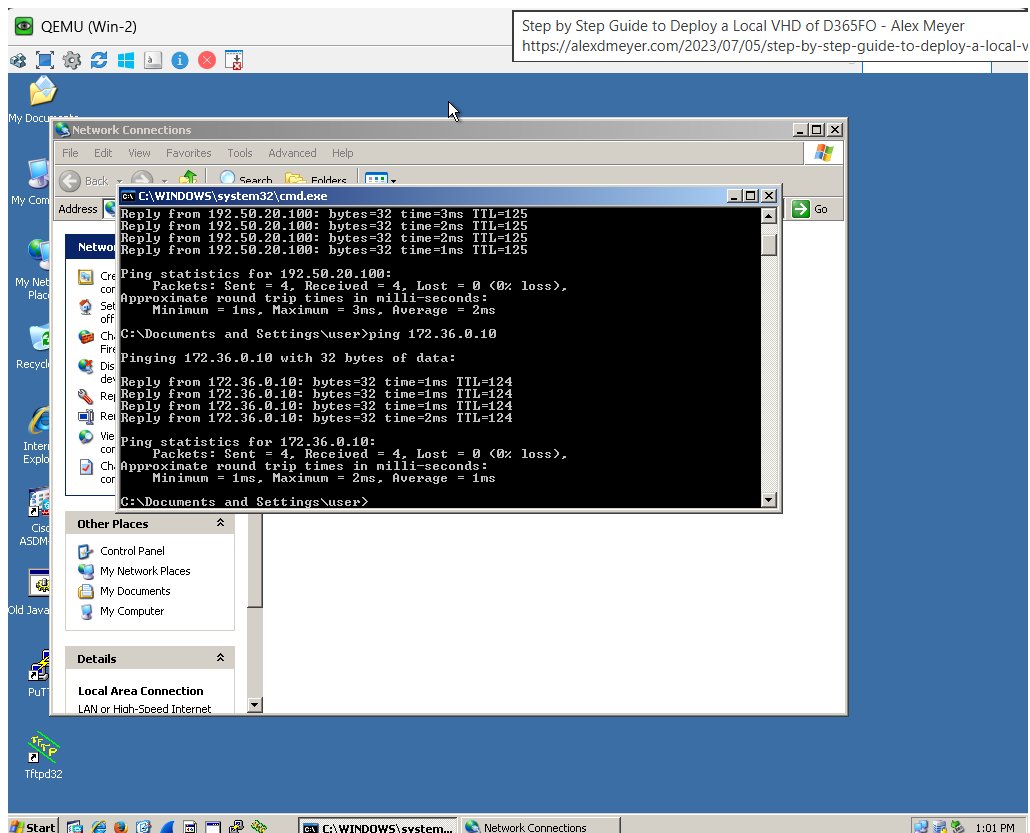
Win-1 additional ping verification



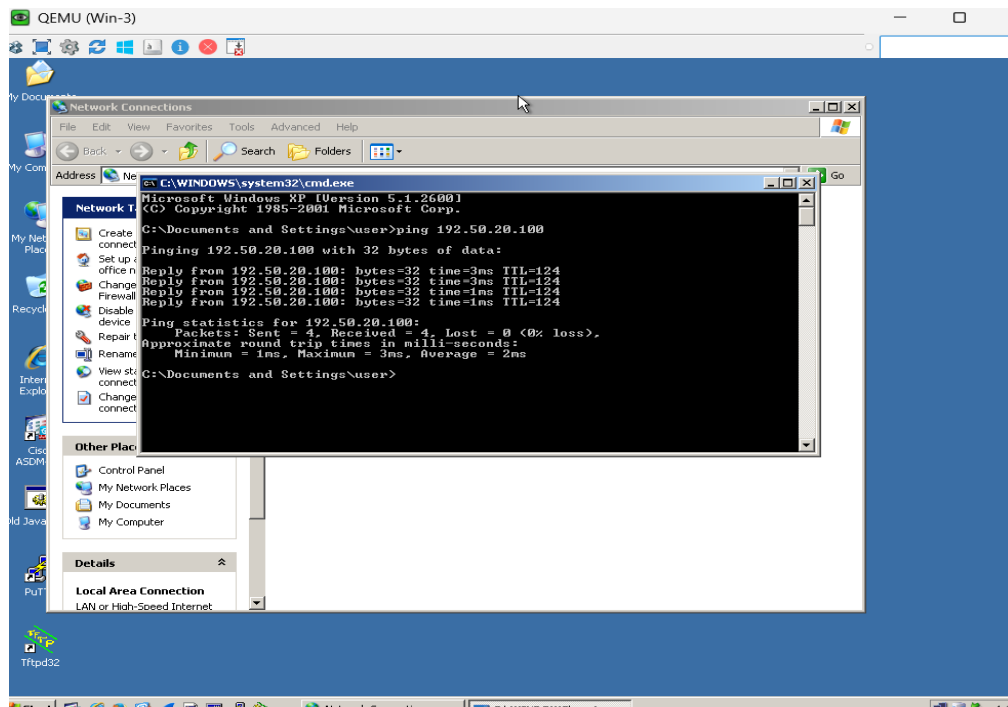
Win-2 ping verification



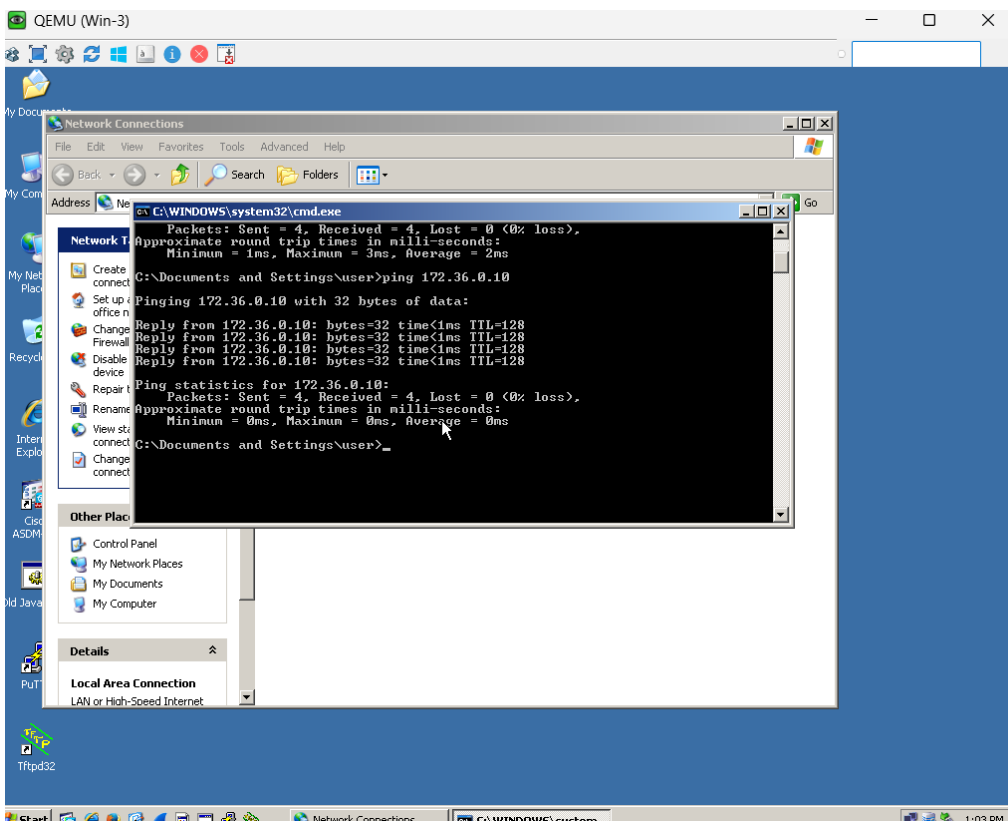
Win-2 additional ping verification



Win-3 ping verification



Win-3 additional ping verification



Final Checklist

1. Configure Win-1 as 192.50.20.100/24 with gateway 192.50.20.1.
2. Configure Win-2 as 223.0.50.10/24 with gateway 223.0.50.1.
3. Configure Win-3 as 172.36.0.10/16 with gateway 172.36.0.1.
4. Configure every router interface with the full IP address, not the short topology label.
5. Use no shutdown on each required interface.
6. Verify with show ip interface brief.
7. Verify routes with show ip route.
8. Perform ping tests between the end hosts.
9. Confirm successful tests show 0% packet loss.